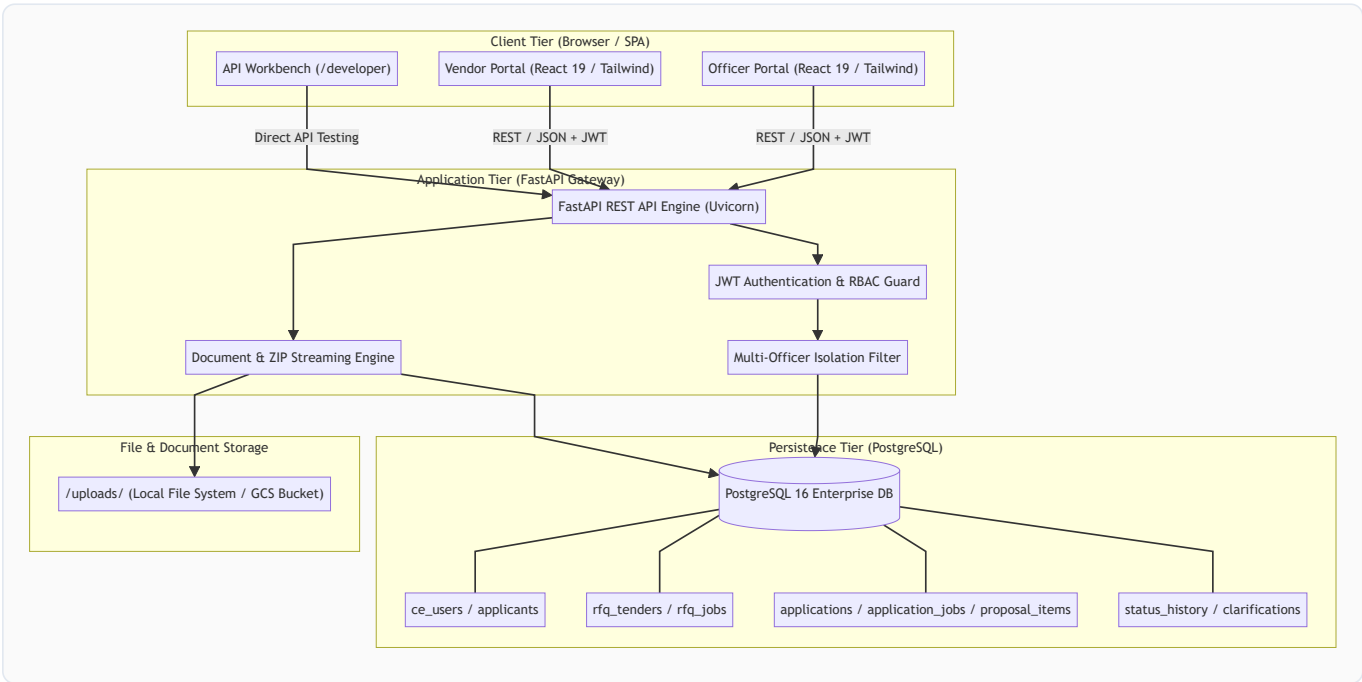


# RFQ Portal — System Architecture, Relational Database Design & REST API Specifications

Amaravati Growth & Infrastructure Corporation Limited (AGIC)

## 1. Executive Summary & System Overview

The **RFQ Portal** is an enterprise procurement and quotation management platform engineered for **Amaravati Growth & Infrastructure Corporation Limited (AGIC)**. The platform streamlines the procurement lifecycle from **RFQ issuance**, **itemized item breakdowns**, **vendor eligibility filtering**, **quotation bidding** (Annexure I, II, III), through to **multi-officer quotation evaluation** and **document packaging**.



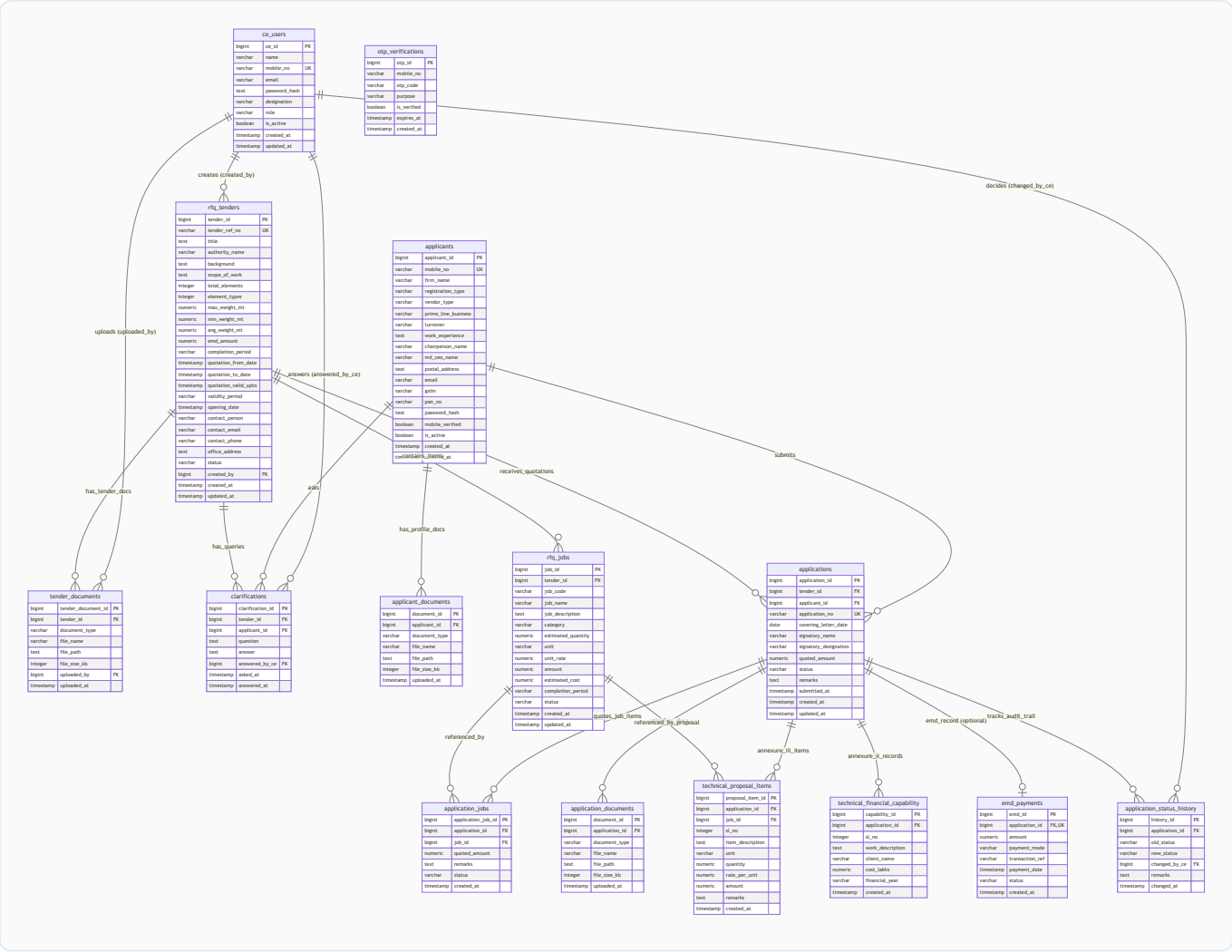
## 2. Technology Stack

| Layer                 | Component / Library | Version   | Purpose                                                     |
|-----------------------|---------------------|-----------|-------------------------------------------------------------|
| Frontend Framework    | React               | 19.x      | Component-based reactive UI rendering                       |
| Build & Tooling       | Vite                | 5.x / 6.x | Lightning-fast HMR and optimized production bundling        |
| Styling & Design      | Tailwind CSS        | 3.4.x     | Utility-first responsive CSS styling with custom AGIC theme |
| Icons & Visuals       | Lucide React        | Latest    | Clean, accessible SVG vector iconography                    |
| Interactive FX        | Canvas Confetti     | Latest    | Celebration animations on successful quotation submission   |
| Backend API Framework | FastAPI             | 0.110+    | High-performance asynchronous Python REST API framework     |
| ASGI Web Server       | Uvicorn             | 0.29+     | Asynchronous server implementation for ASGI applications    |
| ORM & Database Engine | SQLAlchemy          | 2.0+      | Object Relational Mapper with declarative schema modeling   |
| Data Validation       | Pydantic            | v2.x      | Request/Response data parsing, serialization, and typing    |

| Layer                   | Component / Library                                   | Version  | Purpose                                                      |
|-------------------------|-------------------------------------------------------|----------|--------------------------------------------------------------|
| Security & Cryptography | Passlib (Bcrypt) / PyJWT                              | Latest   | SHA-256 / Bcrypt password hashing & JWT token issuance       |
| Database Engine         | PostgreSQL                                            | 16.x     | High-throughput ACID relational database                     |
| File Archiving          | Python <code>zipfile</code> + <code>io.BytesIO</code> | Built-in | Dynamic on-the-fly ZIP generation for multi-document exports |

### 3. Relational Database Architecture

#### 3.1 Entity-Relationship (ER) Diagram



### 4. Comprehensive 15-Table Data Dictionary

Table 1: `ce_users`

Stores credentials, designations, and administrative permissions for Chief Engineers and Officers.

- **Primary Key:** `ce_id` (BigInteger)
- **Indexes:** `ce_id` (PK), `mobile_no` (Unique)

| Column Name        | Data Type           | Nullable | Default        | Constraints / Foreign Keys | Description               |
|--------------------|---------------------|----------|----------------|----------------------------|---------------------------|
| <code>ce_id</code> | <code>BIGINT</code> | NO       | Auto Increment | <code>PRIMARY KEY</code>   | Unique Officer Identifier |

| Column Name   | Data Type    | Nullable | Default          | Constraints / Foreign Keys                    | Description                                     |
|---------------|--------------|----------|------------------|-----------------------------------------------|-------------------------------------------------|
| name          | VARCHAR(150) | NO       | None             | None                                          | Officer's Full Name (e.g. Er. K. V. Ramanathan) |
| mobile_no     | VARCHAR(15)  | NO       | None             | UNIQUE , Indexed                              | Official Login Mobile Number                    |
| email         | VARCHAR(150) | YES      | None             | None                                          | Official Email Address                          |
| password_hash | TEXT         | NO       | None             | None                                          | Bcrypt password hash                            |
| designation   | VARCHAR(100) | YES      | 'Chief Engineer' | None                                          | Official Designation (CE, SE, EE)               |
| role          | VARCHAR(20)  | NO       | 'CE'             | CHECK(role IN ('CE', 'ADMIN', 'SUPER_ADMIN')) | Role-based authorization tier                   |
| is_active     | BOOLEAN      | NO       | TRUE             | None                                          | Account active status flag                      |
| created_at    | TIMESTAMP    | NO       | NOW()            | None                                          | Record creation timestamp                       |
| updated_at    | TIMESTAMP    | NO       | NOW()            | None                                          | Record update timestamp                         |

**Table 2:** applicants

Stores registered vendor/contractor profiles, eligibility classifications, and contact details.

- **Primary Key:** applicant\_id (BigInteger)
- **Indexes:** applicant\_id (PK), mobile\_no (Unique)

| Column Name         | Data Type    | Nullable | Default        | Constraints / Foreign Keys | Description                                                                              |
|---------------------|--------------|----------|----------------|----------------------------|------------------------------------------------------------------------------------------|
| applicant_id        | BIGINT       | NO       | Auto Increment | PRIMARY KEY                | Unique Vendor Identifier                                                                 |
| mobile_no           | VARCHAR(15)  | NO       | None           | UNIQUE , Indexed           | Primary login mobile number                                                              |
| firm_name           | VARCHAR(255) | NO       | None           | None                       | Legal registered entity/business name                                                    |
| registration_type   | VARCHAR(100) | YES      | None           | None                       | Registration class                                                                       |
| vendor_type         | VARCHAR(100) | YES      | None           | None                       | Eligibility Type: Manufacturer , Authorised Dealer , Authorised Distributor , Contractor |
| prime_line_business | VARCHAR(255) | YES      | None           | None                       | Core business focus / manufacturing domain                                               |
| turnover            | VARCHAR(100) | YES      | None           | None                       | Annual financial turnover summary                                                        |
| work_experience     | TEXT         | YES      | None           | None                       | Summary of past structural/supply executions                                             |
| chairperson_name    | VARCHAR(150) | YES      | None           | None                       | Name of Executive Chairperson / Director                                                 |
| md_ceo_name         | VARCHAR(150) | YES      | None           | None                       | Name of Managing Director / CEO                                                          |
| postal_address      | TEXT         | YES      | None           | None                       | Official registered office address                                                       |
| email               | VARCHAR(150) | YES      | None           | None                       | Official correspondence email                                                            |
| gsttin              | VARCHAR(20)  | YES      | None           | None                       | 15-digit GSTIN Registration Number                                                       |
| pan_no              | VARCHAR(20)  | YES      | None           | None                       | 10-character Income Tax PAN Number                                                       |
| password_hash       | TEXT         | YES      | None           | None                       | Bcrypt hash for password login                                                           |
| mobile_verified     | BOOLEAN      | NO       | FALSE          | None                       | Flag verifying OTP mobile validation                                                     |
| is_active           | BOOLEAN      | NO       | TRUE           | None                       | Account status flag                                                                      |
| created_at          | TIMESTAMP    | NO       | NOW()          | None                       | Registration timestamp                                                                   |
| updated_at          | TIMESTAMP    | NO       | NOW()          | None                       | Profile update timestamp                                                                 |

**Table 3:** `applicant_documents`

Stores permanent master profile documents uploaded by the vendor during registration.

- **Primary Key:** `document_id` (BigInteger)
- **Indexes:** `document_id` (PK), `applicant_id` (Index)

| Column Name                | Data Type                 | Nullable | Default                         | Constraints / Foreign Keys                     | Description                                                                                               |
|----------------------------|---------------------------|----------|---------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <code>document_id</code>   | <code>BIGINT</code>       | NO       | Auto Increment                  | <code>PRIMARY KEY</code>                       | Unique Document Identifier                                                                                |
| <code>applicant_id</code>  | <code>BIGINT</code>       | NO       | None                            | <code>FK -&gt; applicants(applicant_id)</code> | Vendor owner reference                                                                                    |
| <code>document_type</code> | <code>VARCHAR(50)</code>  | NO       | <code>'REGISTRATION_DOC'</code> | None                                           | E.g. <code>GST_CERT</code> , <code>PAN_CARD</code> , <code>AUDIT_REPORT</code> , <code>DEALER_AUTH</code> |
| <code>file_name</code>     | <code>VARCHAR(255)</code> | NO       | None                            | None                                           | Original upload file name                                                                                 |
| <code>file_path</code>     | <code>TEXT</code>         | NO       | None                            | None                                           | Relative server storage path ( <code>/uploads/...</code> )                                                |
| <code>file_size_kb</code>  | <code>INTEGER</code>      | YES      | None                            | None                                           | Size of the document in kilobytes                                                                         |
| <code>uploaded_at</code>   | <code>TIMESTAMP</code>    | NO       | <code>NOW()</code>              | None                                           | Timestamp of upload                                                                                       |

**Table 4:** `otp_verifications`

Manages transient 6-digit One-Time-Password tokens for secure SMS login and registration.

- **Primary Key:** `otp_id` (BigInteger)
- **Indexes:** `otp_id` (PK), `mobile_no` (Index)

| Column Name              | Data Type                | Nullable | Default              | Constraints / Foreign Keys | Description                                                                       |
|--------------------------|--------------------------|----------|----------------------|----------------------------|-----------------------------------------------------------------------------------|
| <code>otp_id</code>      | <code>BIGINT</code>      | NO       | Auto Increment       | <code>PRIMARY KEY</code>   | Unique OTP transaction ID                                                         |
| <code>mobile_no</code>   | <code>VARCHAR(15)</code> | NO       | None                 | Indexed                    | Destination mobile number                                                         |
| <code>otp_code</code>    | <code>VARCHAR(10)</code> | NO       | None                 | None                       | Generated 6-digit verification code                                               |
| <code>purpose</code>     | <code>VARCHAR(20)</code> | NO       | <code>'LOGIN'</code> | None                       | Purpose: <code>LOGIN</code> , <code>REGISTER</code> , <code>PASSWORD_RESET</code> |
| <code>is_verified</code> | <code>BOOLEAN</code>     | NO       | <code>FALSE</code>   | None                       | True once successfully consumed                                                   |
| <code>expires_at</code>  | <code>TIMESTAMP</code>   | NO       | None                 | None                       | Expiry cutoff (typically +10 minutes)                                             |
| <code>created_at</code>  | <code>TIMESTAMP</code>   | NO       | <code>NOW()</code>   | None                       | Generation timestamp                                                              |

**Table 5:** `rfq_tenders`

The core procurement request table containing RFQ terms, schedules, specifications, and officer ownership.

- **Primary Key:** `tender_id` (BigInteger)
- **Indexes:** `tender_id` (PK), `tender_ref_no` (Unique), `status` (Index), `created_by` (Index)

| Column Name                 | Data Type                 | Nullable | Default                            | Constraints / Foreign Keys    | Description                                                    |
|-----------------------------|---------------------------|----------|------------------------------------|-------------------------------|----------------------------------------------------------------|
| <code>tender_id</code>      | <code>BIGINT</code>       | NO       | Auto Increment                     | <code>PRIMARY KEY</code>      | Unique RFQ Identifier                                          |
| <code>tender_ref_no</code>  | <code>VARCHAR(50)</code>  | YES      | None                               | <code>UNIQUE</code> , Indexed | Official Reference No (e.g. <code>AGIC/CE/2026/RFQ-01</code> ) |
| <code>title</code>          | <code>TEXT</code>         | NO       | None                               | None                          | Subject / Title of the RFQ                                     |
| <code>authority_name</code> | <code>VARCHAR(255)</code> | NO       | <code>'Amaravati Growth...'</code> | None                          | Procuring Authority Name                                       |

| Column Name          | Data Type     | Nullable | Default | Constraints / Foreign Keys                         | Description                                        |
|----------------------|---------------|----------|---------|----------------------------------------------------|----------------------------------------------------|
| background           | TEXT          | YES      | None    | None                                               | Background & Project Context                       |
| scope_of_work        | TEXT          | YES      | None    | None                                               | Scope of Work, Technical Specs                     |
| total_elements       | INTEGER       | YES      | None    | None                                               | Total precast / structural elements                |
| element_types        | INTEGER       | YES      | None    | None                                               | Number of distinct item types                      |
| max_weight_mt        | NUMERIC(8,2)  | YES      | None    | None                                               | Maximum element weight in Metric Tons              |
| min_weight_mt        | NUMERIC(8,2)  | YES      | None    | None                                               | Minimum element weight in Metric Tons              |
| avg_weight_mt        | NUMERIC(8,2)  | YES      | None    | None                                               | Average element weight in Metric Tons              |
| emd_amount           | NUMERIC(12,2) | NO       | 0.00    | None                                               | EMD required (INR 0.00 for pure RFQ)               |
| completion_period    | VARCHAR(50)   | YES      | None    | None                                               | Execution timeline (e.g. 12 Months )               |
| quotation_from_date  | TIMESTAMP     | NO       | None    | None                                               | Quotation window start datetime                    |
| quotation_to_date    | TIMESTAMP     | NO       | None    | None                                               | Quotation window closing datetime                  |
| quotation_valid_upto | TIMESTAMP     | YES      | None    | None                                               | Exact date up to which quotation must remain valid |
| validity_period      | VARCHAR(100)  | YES      | None    | None                                               | Textual validity descriptor (e.g. 90 Days )        |
| opening_date         | TIMESTAMP     | YES      | None    | None                                               | Date & time for bid opening                        |
| contact_person       | VARCHAR(150)  | YES      | None    | None                                               | Nodal Officer contact name                         |
| contact_email        | VARCHAR(150)  | YES      | None    | None                                               | Nodal Officer email address                        |
| contact_phone        | VARCHAR(20)   | YES      | None    | None                                               | Nodal Officer telephone                            |
| office_address       | TEXT          | YES      | None    | None                                               | Official communication office address              |
| status               | VARCHAR(20)   | NO       | 'DRAFT' | Indexed ( DRAFT , PUBLISHED , CLOSED , CANCELLED ) | Lifecycle status                                   |
| created_by           | BIGINT        | NO       | None    | FK -> ce_users(ce_id) , Indexed                    | Officer ID who owns & manages this RFQ             |
| created_at           | TIMESTAMP     | NO       | NOW()   | None                                               | Creation timestamp                                 |
| updated_at           | TIMESTAMP     | NO       | NOW()   | None                                               | Last update timestamp                              |

**Table 6:** rfq\_jobs

Itemized BOQ / line items associated with an RFQ.

- **Primary Key:** job\_id (BigInteger)
- **Indexes:** job\_id (PK), tender\_id (Index)

| Column Name     | Data Type    | Nullable | Default        | Constraints / Foreign Keys   | Description                       |
|-----------------|--------------|----------|----------------|------------------------------|-----------------------------------|
| job_id          | BIGINT       | NO       | Auto Increment | PRIMARY KEY                  | Unique Item Identifier            |
| tender_id       | BIGINT       | NO       | None           | FK -> rfq_tenders(tender_id) | Parent RFQ reference              |
| job_code        | VARCHAR(30)  | YES      | None           | None                         | Item Code (e.g. ITEM-01 , PKG-A ) |
| job_name        | VARCHAR(255) | NO       | None           | None                         | Item Name / Nomenclature          |
| job_description | TEXT         | YES      | None           | None                         | Detailed technical specification  |

| Column Name        | Data Type     | Nullable | Default       | Constraints / Foreign Keys | Description                                                 |
|--------------------|---------------|----------|---------------|----------------------------|-------------------------------------------------------------|
| category           | VARCHAR(100)  | YES      | 'Supply Item' | None                       | Type: <code>Supply Item</code> or <code>Only Rate</code>    |
| estimated_quantity | NUMERIC(12,2) | YES      | None          | None                       | Required Quantity                                           |
| unit               | VARCHAR(30)   | YES      | 'NOS'         | None                       | Measurement unit (NOS, MT, SQM, RMT, CUM)                   |
| unit_rate          | NUMERIC(16,2) | YES      | None          | None                       | Departmental reference rate (optional)                      |
| amount             | NUMERIC(16,2) | YES      | None          | None                       | Total estimated amount ( <code>qty * unit_rate</code> )     |
| estimated_cost     | NUMERIC(16,2) | YES      | None          | None                       | Estimated cost baseline                                     |
| completion_period  | VARCHAR(50)   | YES      | None          | None                       | Item-level completion timeline                              |
| status             | VARCHAR(20)   | NO       | 'ACTIVE'      | None                       | Item status ( <code>ACTIVE</code> , <code>INACTIVE</code> ) |
| created_at         | TIMESTAMP     | NO       | NOW()         | None                       | Creation timestamp                                          |
| updated_at         | TIMESTAMP     | NO       | NOW()         | None                       | Update timestamp                                            |

**Table 7:** `tender_documents`

Official RFQ notices, technical drawings, and specifications uploaded by the Chief Engineer.

- **Primary Key:** `tender_document_id` (BigInteger)
- **Indexes:** `tender_document_id` (PK), `tender_id` (Index)

| Column Name        | Data Type    | Nullable | Default        | Constraints / Foreign Keys                | Description                                                                      |
|--------------------|--------------|----------|----------------|-------------------------------------------|----------------------------------------------------------------------------------|
| tender_document_id | BIGINT       | NO       | Auto Increment | PRIMARY KEY                               | Unique Document Identifier                                                       |
| tender_id          | BIGINT       | NO       | None           | FK -> <code>rfq_tenders(tender_id)</code> | Parent RFQ reference                                                             |
| document_type      | VARCHAR(50)  | NO       | 'RFQ_DOCUMENT' | None                                      | E.g. <code>RFQ_NOTICE</code> , <code>DRAWING</code> , <code>SPECIFICATION</code> |
| file_name          | VARCHAR(255) | NO       | None           | None                                      | Original document file name                                                      |
| file_path          | TEXT         | NO       | None           | None                                      | Relative server path ( <code>/uploads/...</code> )                               |
| file_size_kb       | INTEGER      | YES      | None           | None                                      | File size in KB                                                                  |
| uploaded_by        | BIGINT       | YES      | None           | FK -> <code>ce_users(ce_id)</code>        | Uploading Officer ID                                                             |
| uploaded_at        | TIMESTAMP    | NO       | NOW()          | None                                      | Upload timestamp                                                                 |

**Table 8:** `applications`

Master table for quotation submissions received from eligible vendors against published RFQs.

- **Primary Key:** `application_id` (BigInteger)
- **Indexes:** `application_id` (PK), `application_no` (Unique), `tender_id` (Index), `applicant_id` (Index), `status` (Index)

| Column Name    | Data Type | Nullable | Default        | Constraints / Foreign Keys                  | Description                    |
|----------------|-----------|----------|----------------|---------------------------------------------|--------------------------------|
| application_id | BIGINT    | NO       | Auto Increment | PRIMARY KEY                                 | Unique Quotation Submission ID |
| tender_id      | BIGINT    | NO       | None           | FK -> <code>rfq_tenders(tender_id)</code>   | RFQ being quoted               |
| applicant_id   | BIGINT    | NO       | None           | FK -> <code>applicants(applicant_id)</code> | Bidding Vendor reference       |

| Column Name           | Data Type     | Nullable | Default     | Constraints / Foreign Keys                                             | Description                                              |
|-----------------------|---------------|----------|-------------|------------------------------------------------------------------------|----------------------------------------------------------|
| application_no        | VARCHAR(50)   | YES      | None        | UNIQUE , Indexed                                                       | Auto-generated Ref No ( APP-{tender}-{app}-{timestamp} ) |
| covering_letter_date  | DATE          | YES      | None        | None                                                                   | Date of Annexure I Covering Letter                       |
| signatory_name        | VARCHAR(150)  | YES      | None        | None                                                                   | Authorized signatory full name                           |
| signatory_designation | VARCHAR(150)  | YES      | None        | None                                                                   | Signatory title / designation                            |
| quoted_amount         | NUMERIC(16,2) | YES      | None        | None                                                                   | Grand Total Quoted Amount (INR)                          |
| status                | VARCHAR(30)   | NO       | 'SUBMITTED' | Indexed ( SUBMITTED , UNDER_REVIEW , ACCEPTED , REJECTED , WITHDRAWN ) | Evaluation status                                        |
| remarks               | TEXT          | YES      | None        | None                                                                   | Evaluation comments or rejection reasons                 |
| submitted_at          | TIMESTAMP     | NO       | NOW()       | None                                                                   | Quotation submission timestamp                           |
| created_at            | TIMESTAMP     | NO       | NOW()       | None                                                                   | Creation timestamp                                       |
| updated_at            | TIMESTAMP     | NO       | NOW()       | None                                                                   | Last status update timestamp                             |

**Table 9:** application\_jobs

Maps which specific RFQ items were included in a vendor's quotation, along with package-level quotes.

- **Primary Key:** application\_job\_id (BigInteger)
- **Indexes:** application\_job\_id (PK), application\_id (Index), job\_id (Index)

| Column Name        | Data Type     | Nullable | Default        | Constraints / Foreign Keys         | Description                                |
|--------------------|---------------|----------|----------------|------------------------------------|--------------------------------------------|
| application_job_id | BIGINT        | NO       | Auto Increment | PRIMARY KEY                        | Unique Mapping Identifier                  |
| application_id     | BIGINT        | NO       | None           | FK -> applications(application_id) | Parent Quotation submission                |
| job_id             | BIGINT        | NO       | None           | FK -> rfq_jobs(job_id)             | Quoted RFQ Item reference                  |
| quoted_amount      | NUMERIC(16,2) | YES      | None           | None                               | Item-level total quoted price              |
| remarks            | TEXT          | YES      | None           | None                               | Vendor's item remarks / delivery timeline  |
| status             | VARCHAR(20)   | NO       | 'SUBMITTED'    | None                               | Status ( SUBMITTED , ACCEPTED , REJECTED ) |
| created_at         | TIMESTAMP     | NO       | NOW()          | None                               | Creation timestamp                         |

**Table 10:** application\_documents

Stores supporting documents uploaded by the vendor during quotation submission (e.g. Annexure-I signed letter, Annexure-III price schedule, power of attorney).

- **Primary Key:** document\_id (BigInteger)
- **Indexes:** document\_id (PK), application\_id (Index)

| Column Name | Data Type | Nullable | Default        | Constraints / Foreign Keys | Description                |
|-------------|-----------|----------|----------------|----------------------------|----------------------------|
| document_id | BIGINT    | NO       | Auto Increment | PRIMARY KEY                | Unique Document Identifier |

| Column Name    | Data Type    | Nullable | Default | Constraints / Foreign Keys         | Description                                     |
|----------------|--------------|----------|---------|------------------------------------|-------------------------------------------------|
| application_id | BIGINT       | NO       | None    | FK -> applications(application_id) | Parent quotation submission                     |
| document_type  | VARCHAR(50)  | NO       | None    | None                               | E.g. ANNEXURE_I_LETTER , ANNEXURE_III_BOQ , POA |
| file_name      | VARCHAR(255) | NO       | None    | None                               | Original upload file name                       |
| file_path      | TEXT         | NO       | None    | None                               | Storage destination ( /uploads/... )            |
| file_size_kb   | INTEGER      | YES      | None    | None                               | Size in KB                                      |
| uploaded_at    | TIMESTAMP    | NO       | NOW()   | None                               | Upload timestamp                                |

**Table 11:** technical\_financial\_capability

Stores **Annexure II** past execution experience and credentials submitted with the quotation.

- **Primary Key:** capability\_id (BigInteger)
- **Indexes:** capability\_id (PK), application\_id (Index)

| Column Name      | Data Type     | Nullable | Default        | Constraints / Foreign Keys         | Description                                |
|------------------|---------------|----------|----------------|------------------------------------|--------------------------------------------|
| capability_id    | BIGINT        | NO       | Auto Increment | PRIMARY KEY                        | Unique Record Identifier                   |
| application_id   | BIGINT        | NO       | None           | FK -> applications(application_id) | Parent quotation submission                |
| sl_no            | INTEGER       | NO       | None           | None                               | Serial Number (1, 2, 3...)                 |
| work_description | TEXT          | NO       | None           | None                               | Project name and description               |
| client_name      | VARCHAR(255)  | YES      | None           | None                               | Government / Corporate Client Name         |
| cost_lakhs       | NUMERIC(14,2) | YES      | None           | None                               | Project cost executed (in Lakhs INR)       |
| financial_year   | VARCHAR(20)   | YES      | None           | None                               | Financial Year completed (e.g. 2024-2025 ) |
| created_at       | TIMESTAMP     | NO       | NOW()          | None                               | Record creation timestamp                  |

**Table 12:** technical\_proposal\_items

Stores **Annexure III** detailed Bill of Quantities (BOQ) itemized financial quotes submitted by the vendor.

- **Primary Key:** proposal\_item\_id (BigInteger)
- **Indexes:** proposal\_item\_id (PK), application\_id (Index), job\_id (Index)

| Column Name      | Data Type     | Nullable | Default        | Constraints / Foreign Keys         | Description                        |
|------------------|---------------|----------|----------------|------------------------------------|------------------------------------|
| proposal_item_id | BIGINT        | NO       | Auto Increment | PRIMARY KEY                        | Unique Proposal Item ID            |
| application_id   | BIGINT        | NO       | None           | FK -> applications(application_id) | Parent quotation submission        |
| job_id           | BIGINT        | YES      | None           | FK -> rfq_jobs(job_id)             | Optional link to specific RFQ Job  |
| sl_no            | INTEGER       | NO       | None           | None                               | Line item serial number            |
| item_description | TEXT          | NO       | None           | None                               | Description of goods / works       |
| unit             | VARCHAR(30)   | YES      | None           | None                               | Unit of measurement (NOS, MT, SQM) |
| quantity         | NUMERIC(14,2) | YES      | None           | None                               | Proposed item quantity             |

| Column Name   | Data Type     | Nullable | Default | Constraints / Foreign Keys | Description                                        |
|---------------|---------------|----------|---------|----------------------------|----------------------------------------------------|
| rate_per_unit | NUMERIC(14,2) | YES      | None    | None                       | Quoted Unit Rate (INR)                             |
| amount        | NUMERIC(16,2) | YES      | None    | None                       | Total line item amount ( <code>qty * rate</code> ) |
| remarks       | TEXT          | YES      | None    | None                       | Specifications / Manufacturer brand remarks        |
| created_at    | TIMESTAMP     | NO       | NOW()   | None                       | Record timestamp                                   |

**Table 13:** emd\_payments

Tracks Earnest Money Deposit transactions (maintained for legacy compatibility or special security deposits).

- **Primary Key:** emd\_id (BigInteger)
- **Indexes:** emd\_id (PK), application\_id (Unique Index)

| Column Name     | Data Type     | Nullable | Default        | Constraints / Foreign Keys                     | Description                                     |
|-----------------|---------------|----------|----------------|------------------------------------------------|-------------------------------------------------|
| emd_id          | BIGINT        | NO       | Auto Increment | PRIMARY KEY                                    | Unique EMD record identifier                    |
| application_id  | BIGINT        | NO       | None           | FK -> applications(application_id) ,<br>UNIQUE | One-to-one quotation link                       |
| amount          | NUMERIC(12,2) | NO       | None           | None                                           | Paid EMD amount                                 |
| payment_mode    | VARCHAR(30)   | YES      | None           | None                                           | Mode: ONLINE_PORTAL , NEFT ,<br>RTGS , EXEMPTED |
| transaction_ref | VARCHAR(100)  | YES      | None           | None                                           | Bank UTR / Transaction reference ID             |
| payment_date    | TIMESTAMP     | YES      | None           | None                                           | Timestamp of transaction                        |
| status          | VARCHAR(20)   | NO       | 'PENDING'      | None                                           | Status ( SUCCESS , PENDING ,<br>EXEMPTED )      |
| created_at      | TIMESTAMP     | NO       | NOW()          | None                                           | Record timestamp                                |

**Table 14:** application\_status\_history

Full audit trail tracking every evaluation state transition, remarks, and Officer identity.

- **Primary Key:** history\_id (BigInteger)
- **Indexes:** history\_id (PK), application\_id (Index)

| Column Name    | Data Type   | Nullable | Default        | Constraints / Foreign Keys         | Description                                              |
|----------------|-------------|----------|----------------|------------------------------------|----------------------------------------------------------|
| history_id     | BIGINT      | NO       | Auto Increment | PRIMARY KEY                        | Unique Audit Record Identifier                           |
| application_id | BIGINT      | NO       | None           | FK -> applications(application_id) | Quotation reference                                      |
| old_status     | VARCHAR(30) | YES      | None           | None                               | Status before transition                                 |
| new_status     | VARCHAR(30) | NO       | None           | None                               | New status set ( UNDER_REVIEW , ACCEPTED ,<br>REJECTED ) |
| changed_by_ce  | BIGINT      | YES      | None           | FK -> ce_users(ce_id)              | Officer ID who performed the action                      |
| remarks        | TEXT        | YES      | None           | None                               | Recorded justification or audit notes                    |
| changed_at     | TIMESTAMP   | NO       | NOW()          | None                               | Timestamp of change                                      |

**Table 15:** clarifications

Interactive pre-bid technical query and clarification system between Vendors and Chief Engineers.

- **Primary Key:** clarification\_id (BigInteger)
- **Indexes:** clarification\_id (PK), tender\_id (Index)

| Column Name      | Data Type | Nullable | Default        | Constraints / Foreign Keys     | Description                        |
|------------------|-----------|----------|----------------|--------------------------------|------------------------------------|
| clarification_id | BIGINT    | NO       | Auto Increment | PRIMARY KEY                    | Unique Query Identifier            |
| tender_id        | BIGINT    | NO       | None           | FK -> rfq_tenders(tender_id)   | Target RFQ reference               |
| applicant_id     | BIGINT    | YES      | None           | FK -> applicants(applicant_id) | Vendor who posed the question      |
| question         | TEXT      | NO       | None           | None                           | Technical or commercial query text |
| answer           | TEXT      | YES      | None           | None                           | Official response by the Officer   |
| answered_by_ce   | BIGINT    | YES      | None           | FK -> ce_users(ce_id)          | Answering Officer ID               |
| asked_at         | TIMESTAMP | NO       | NOW()          | None                           | Submission timestamp               |
| answered_at      | TIMESTAMP | YES      | None           | None                           | Response timestamp                 |

## 5. REST API Specifications

**Base URL:** http://localhost:8000/api

### 5.1 Authentication & Profile Endpoints ( /api/auth )

| Method | Endpoint                                     | Authorization | Description                                        | Request Body                                                            | Response Codes                                              |
|--------|----------------------------------------------|---------------|----------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------|
| POST   | /api/auth/applicant/request-otp              | Public        | Generate 6-digit OTP for Vendor login/registration | { "mobile_no": "9111222333", "purpose": "LOGIN" }                       | 200 OK , 400 Bad Request                                    |
| POST   | /api/auth/applicant/verify-otp               | Public        | Verify OTP and issue Vendor JWT token              | { "mobile_no": "9111222333", "otp_code": "123456", "purpose": "LOGIN" } | 200 OK (TokenResponse) , 400 Bad Request , 404 Not Found    |
| POST   | /api/auth/applicant/register                 | Public        | Register new Vendor profile and issue JWT          | ApplicantRegisterRequest (JSON)                                         | 200 OK (TokenResponse) , 400 Conflict                       |
| POST   | /api/auth/applicant/login                    | Public        | Password login for Vendor                          | { "mobile_no": "...", "password": "..." }                               | 200 OK (TokenResponse) , 401 Unauthorized                   |
| POST   | /api/auth/applicant/{applicant_id}/documents | Vendor        | Upload master profile documents                    | multipart/form-data ( file , document_type )                            | 200 OK , 404 Not Found                                      |
| POST   | /api/auth/ce/login                           | Public        | Login for Chief Engineers and Officers             | { "mobile_no": "9876543210", "password": "..." }                        | 200 OK (TokenResponse) , 401 Unauthorized , 403 Deactivated |
| GET    | /api/auth/me                                 | Bearer JWT    | Fetch active session profile and role permissions  | Header: Authorization: Bearer <token>                                   | 200 OK (Profile Model) , 401 Unauthorized                   |

## 5.2 RFQ Management Endpoints ( /api/tenders )

| Method | Endpoint                               | Authorization  | Description                                                                                     | Request Body / Query Params                                                         | Response Codes                                     |
|--------|----------------------------------------|----------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------|
| GET    | /api/tenders                           | Public / Auth  | List all RFQs with item summaries and window status                                             | Query: <code>status_filter</code> , <code>created_by</code>                         | 200 OK (List[TenderOut])                           |
| GET    | /api/tenders/{tender_id}               | Public / Auth  | Get comprehensive RFQ details with items and documents                                          | Path: <code>tender_id</code> (int)                                                  | 200 OK (TenderOut) , 404 Not Found                 |
| POST   | /api/tenders                           | Officer ( CE ) | Create and publish a new RFQ with items                                                         | <code>TenderCreate</code> (JSON)                                                    | 200 OK (TenderOut) , 403 Forbidden                 |
| PUT    | /api/tenders/{tender_id}               | Officer Owner  | Update RFQ details, dates, items, and validity period                                           | <code>TenderUpdate</code> (JSON)                                                    | 200 OK (TenderOut) , 403 Forbidden , 404 Not Found |
| POST   | /api/tenders/{tender_id}/jobs          | Officer Owner  | Add a single RFQ Item to an existing RFQ                                                        | <code>RFQJobCreate</code> (JSON)                                                    | 200 OK (RFQJobOut) , 403 Forbidden                 |
| DELETE | /api/tenders/{tender_id}/jobs/{job_id} | Officer Owner  | Delete an RFQ Item from an RFQ                                                                  | Path: <code>tender_id</code> , <code>job_id</code>                                  | 200 OK , 403 Forbidden , 404 Not Found             |
| PATCH  | /api/tenders/{tender_id}/status        | Officer Owner  | Transition RFQ status ( <code>PUBLISHED</code> , <code>CLOSED</code> , <code>CANCELLED</code> ) | Query: <code>new_status</code>                                                      | 200 OK , 400 Bad Request , 403 Forbidden           |
| POST   | /api/tenders/{tender_id}/documents     | Officer Owner  | Upload official RFQ specification / drawing                                                     | <code>multipart/form-data</code> ( <code>file</code> , <code>document_type</code> ) | 200 OK , 403 Forbidden                             |

## 5.3 Quotation Submissions & Review Endpoints ( /api/applications )

| Method | Endpoint                                     | Authorization   | Description                                              | Request Body / Query Params                                                         | Response Codes                                                |
|--------|----------------------------------------------|-----------------|----------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------|
| POST   | /api/applications                            | Eligible Vendor | Submit a complete Quotation (Annexure I, II, III & Jobs) | <code>ApplicationSubmitRequest</code> (JSON)                                        | 200 OK (ApplicationOut) , 400 Ineligible / Closed / Duplicate |
| POST   | /api/applications/{application_id}/documents | Vendor Owner    | Upload signed Annexure sheets or supporting files        | <code>multipart/form-data</code> ( <code>file</code> , <code>document_type</code> ) | 200 OK , 403 Forbidden , 404 Not Found                        |
| GET    | /api/applications/my                         | Vendor          | Fetch all quotations submitted by the logged-in vendor   | Header: <code>Authorization: Bearer &lt;token&gt;</code>                            | 200 OK (List[ApplicationOut])                                 |

| Method | Endpoint                                        | Authorization   | Description                                                                                          | Request Body / Query Params                                             | Response Codes                                        |
|--------|-------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------|
| GET    | /api/applications/officer/all                   | Officer ( CE )  | <b>Master Quotations Review Center:</b><br>View quotations strictly for RFQs created by this Officer | Header: Authorization: Bearer <token>                                   | 200 OK (List[ApplicationOut])                         |
| GET    | /api/applications/tender/{tender_id}            | Officer Owner   | List all quotations submitted for a specific RFQ                                                     | Path: tender_id                                                         | 200 OK (List[ApplicationOut]), 403 Forbidden          |
| GET    | /api/applications/{application_id}              | Owner / Officer | Get complete details of a single quotation                                                           | Path: application_id                                                    | 200 OK (ApplicationOut), 403 Forbidden, 404 Not Found |
| GET    | /api/applications/{application_id}/download-all | Owner / Officer | <b>Download All Documents as ZIP:</b><br>Packages all Annexure & Profile docs                        | Query: ?token=<jwt_token> (for browser direct download)                 | 200 OK (Streaming application/zip), 403 Forbidden     |
| PATCH  | /api/applications/{application_id}/status       | Officer Owner   | <b>Approve / Accept / Reject Quotation</b> with audit remarks                                        | {"status": "ACCEPTED", "remarks": "Approved as lowest compliant quote"} | 200 OK, 400 Bad Request, 403 Forbidden                |

#### 5.4 Clarifications & Queries Endpoints ( /api/clarifications )

| Method | Endpoint                                      | Authorization | Description                                      | Request Body                        | Response Codes                           |
|--------|-----------------------------------------------|---------------|--------------------------------------------------|-------------------------------------|------------------------------------------|
| GET    | /api/clarifications/tender/{tender_id}        | Public / Auth | List all public Q&As for a specific RFQ          | Path: tender_id                     | 200 OK (List[ClarificationOut])          |
| POST   | /api/clarifications                           | Vendor        | Post a pre-bid technical clarification question  | {"tender_id": 1, "question": "..."} | 200 OK (ClarificationOut), 404 Not Found |
| POST   | /api/clarifications/{clarification_id}/answer | Officer Owner | Provide official department answer to a question | {"answer": "..."}                   | 200 OK (ClarificationOut), 404 Not Found |

## 5.5 Analytics & Statistics Endpoints ( /api/stats )

| Method | Endpoint             | Authorization  | Description                                                                      | Response Schema                                                                                                                                                                                                                                                           |
|--------|----------------------|----------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET    | /api/stats/ce        | Officer ( CE ) | Aggregated metrics for Officer Dashboard (scoped strictly to Officer's own RFQs) | <code>{"total_tenders": 5, "active_tenders": 4, "total_applications": 12, "under_review_applications": 3, "accepted_applications": 4, "rejected_applications": 1, "total_registered_applicants": 5, "total_quoted_amount": 54200000.0, "total_emd_collected": 0.0}</code> |
| GET    | /api/stats/applicant | Vendor         | Personal bidding counters for Vendor Dashboard                                   | <code>{"my_total_applications": 6, "submitted": 2, "under_review": 1, "accepted": 3}</code>                                                                                                                                                                               |

## 6. Multi-Tenant Security & Isolation Model

The RFQ Portal implements **strict departmental isolation** between different Chief Engineers:

### 1. RFQ Creation & Ownership:

- Each RFQ record stores `created_by = current_user.ce_id`.
- Modifying RFQ metadata, extending dates, updating status, and adding/deleting RFQ items is strictly gated to the creating Officer (`tender.created_by == current_user.id`).

### 2. Master Quotations Review Center Isolation:

- When Officer A accesses the Master Quotations Review Center ( `GET /api/applications/officer/all` ), the backend executes:

```
ce_tenders = db.query(RFQTender.tender_id).filter(RFQTender.created_by == current_user["id"]).all()
tender_ids = [t[0] for t in ce_tenders]
apps = db.query(Application).filter(Application.tender_id.in_(tender_ids)).all()
```

- Officer A **never** sees or receives quotation payloads belonging to RFQs created by Officer B.

### 3. Document ZIP Package Protection:

- The document archive streaming endpoint ( `GET /api/applications/{id}/download-all` ) verifies ownership before opening the file stream:

```
if current_user["role"] in ["CE", "ADMIN"] and current_user["role"] != "SUPER_ADMIN":
    tender = db.query(RFQTender).filter(RFQTender.tender_id == app_obj.tender_id).first()
    if tender and tender.created_by != current_user["id"]:
        raise HTTPException(status_code=403, detail="Access forbidden: You do not have authorization to download quotation docu
```

### 4. Vendor Eligibility Protection:

- Only registered entities under permitted classifications ( `MANUFACTURER` , `AUTHORISED_DEALER` , `AUTHORISED_DISTRIBUTOR` , `CONTRACTOR` ) are permitted to execute `POST /api/applications` . Unregistered or unauthorized roles receive a `400 Bad Request` with an explicit eligibility rejection.

## 7. Configuration & Environment Variables

| Variable Name | Default Value                                        | Description                                               |
|---------------|------------------------------------------------------|-----------------------------------------------------------|
| PROJECT_NAME  | RFQ Portal AGIC                                      | Display name of the application                           |
| DATABASE_URL  | postgresql://postgres:postgres@localhost:5432/rfq_db | PostgreSQL connection DSN                                 |
| SECRET_KEY    | agic-rfq-portal-super-secret-key-2026-production     | Secret key used to sign and verify HMAC-SHA256 JWT tokens |

| Variable Name               | Default Value                                      | Description                                         |
|-----------------------------|----------------------------------------------------|-----------------------------------------------------|
| ALGORITHM                   | HS256                                              | Cryptographic algorithm for JWT                     |
| ACCESS_TOKEN_EXPIRE_MINUTES | 1440 (24 Hours)                                    | Lifetime of session bearer tokens                   |
| UPLOAD_DIR                  | ./uploads                                          | Physical directory for storing document attachments |
| CORS_ORIGINS                | ["http://localhost:5173", "http://localhost:3000"] | Whitelisted frontend origins                        |

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